



Faculty of Engineering and the Built Environment

Department of Electrical Engineering

2026 Course Handout: EEE4114F

Course Name:	Digital Signal Processing
SAQA Credits:	16
Pre-requisites:	EEE3092F OR EEE3094S
Co-requisites:	N/A

Course convenor:	Dr Yaaseen Martin
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Office location:	Menzies 6.12
Consultation hours:	By arrangement (contact via email)
Course lecturer:	Dr Yaaseen Martin and Mr Jarryd Son
Teaching assistant:	Mr Tohlang Rakhapu

Lecture venue:	CENG SEM
Lecture days and time:	Tuesdays and Thursdays, 14:00 – 16:00
Laboratory venue	N/A
Laboratory days and times	N/A

Course objectives

Part A of this course aims to develop an advanced understanding of digital signal processing. Topics include: discrete time signals and systems; the discrete Fourier transform properties and fast algorithms; the z-transform; frequency response from z- plane; FIR and IIR filter design and structures for digital filters. The course includes a specialist component in an applied or advanced signal processing application area.

Part B of the course is an introduction to concepts, terminology, algorithms and techniques of machine learning (ML) for Electrical Engineering students. The course aims to develop an understanding of ML concepts and algorithms, techniques for handling data for ML applications, and to design and implement appropriate ML models for a given application.

Learning outcomes (indicate the level at which GAs are addressed: **B (basic)**, **I (intermediate)**, **A (advanced)**).

Also indicate **Y** or **N** if at Exit Level

Students successfully completing this course will have the following:	Exit level? Y/N	GA1	GA2	GA3	GA4	GA5	GA6	GA7	GA8	GA9	GA10	GA11
A. Knowledge (Information plus Understanding)												
1. Representation of discrete-time signals and systems, both in time and frequency domain	N					A						
2. Discrete-time Fourier transforms, Z-transforms, wavelet and other joint time-frequency transforms, and their application	N					A						

3. How to convert between continuous-time and discrete-time domains	N					A									
4. Using digital signal processing for filtering application	N		A												
5. Supervised, unsupervised and reinforcement learning algorithms, and their application	N					A									
6. Understanding the influence of feature representation on the training performance of ML algorithm	N					A									
7. Interpreting and evaluating results from ML models	N					A									
B. Skills (Application of Knowledge)															
1. Be able to calculate responses (in various different ways) of discrete-time systems to given signal	N	A													
2. Classify discrete-time systems according to their properties	N		A												
3. Convert between different but equivalent representations of linear time-variant systems	N		A												
4. Design and implement FIR filters for specific application	N			A											
5. Be able to develop data pre-processing pipelines to extract and represent features to facilitate efficient training of ML algorithm	N				A										
6. Be able to implement a variety of machine learning algorithms to solve supervised learning tasks	N			A											
7. Identify common machine learning tasks and the appropriate methods to solve them	N		A												
8. Apply common techniques to analyse and evaluate the performance of ML algorithms	N				A										
C. Values and Attitudes															
1. Appreciate the value of fundamental mathematical sciences to the procedural design of complex digital signal processing and machine learning algorithm	N		A												
2. Develop an understanding and application of the design / synthesis procedure	N			A											
3. See advanced applications of digital signal processing, transform theory and machine learning	N		A												

Detailed course content

Upon completion of the course students will be able to calculate responses (in various different ways) of discrete-time systems to given signals; classify discrete-time systems according to their properties (e.g. stability); derive frequency responses for given digital filters; design FIR and IIR filters for desired frequency

and phase responses; be able to select best filter architecture for a specific implementation, and use appropriate windows in spectrum estimation. They will also have basic familiarity with methods in selected areas where signal processing is applied.

Students will be able to perform data analysis and pre-processing procedures commonly used in machine learning applications such as feature selection, feature scaling, encoding; implement supervised learning algorithms, such as, linear and logistic regression, nearest neighbour algorithms and artificial neural networks (including deep neural networks) to solve common classification and regression problems; apply sound evaluation techniques to appropriately interpret model performance; students will also be introduced to the fundamentals of reinforcement learning and unsupervised learning

Test Dates:

Class Test 1	24 th March, 18:00 – 19:00
Class Test 2	7 th May, 18:00 – 19:00

Knowledge areas

Maths Sciences	Natural Sciences	Eng Sciences	Design & Synthesis	Complm Studies
-	-	75	25	-

Learning environment

The course has been designed around four types of activities: (1) lectures, (2) problem sets, (5) computer assignments, and (4) assessments.

These will all be supported by 4 contact sessions per week (45 minutes each) to assist with pacing and provide academic support.

An average week will consist of (at least some of) the following:

1. **Lecture support:** Recent content will be revised, and future content will be introduced. The objective here is to provide a broad picture of the course and the relevance of its content.
2. **Worked examples/Q&A:** Lecture slots will also be used to provide worked examples to consolidate the skills that the course aims to develop. These will ideally be interactive with students providing problems that they (and others) are having difficulty with.
3. **Tutorials:** There are formal problem sets that students can choose to engage with, and the expectation is that students will engage constantly with problems relevant to each section. Some of these may come from past tests and assignments. It is intended that the tutorials provide general guidance on relevant problem solving but also specifically aid in the submission of problem sets.
4. **Computer assignments:** There are multiple DSP and ML computer assignments to consolidate concepts and develop software and numerical programming competence.

Suggested time allocation

Learning Activity	No.	Time in hours	Contact time Multiplier	Total no of hours
Number of lectures	38	0.75	2	57
Number of tutorials	12	1	2	24
Total practical/lab periods	6	2	2	24
Total other contact periods	-	-	-	-
Total assignment non-contact hours		30		30

Assessment hours (Tests, Exam)		5	5	25
Number of weeks the course lasts	13			
Total hours				160

General assessment strategy

Assessment Task	%	The following DP rules apply:
Assignments	-	- Completion and submission of all computer assignments - Completion and submission of the course mini-project
Tutorials	-	
Labs	-	
Project	10	
Tests	20	
Exam	70	
Total	100	

Books/Reading Materials/Notes

- Recommended text: Oppenheim and Schafer, Discrete-Time Signal Processing - Comprehensive set of notes (for each of DSP and ML) given out during the course

Absence: The continuous assessment marks will be adjusted to allow for absence only on the following grounds:

- A medical certificate for absence of 3 or more consecutive days or any day with a class test or exam
- Death of an immediate family member (parent or sibling)
- Pre-arranged absence to represent a University, provincial or national team.

Academic dishonesty: Plagiarism is a very serious offence and usually leads to disciplinary action that could include expulsion from the university. Therefore, recognise the work of others in any submission. Details of referencing methods are widely available on the Web. A non-plagiarism declaration must be submitted with all work submitted for marking.